

Cornetto after party



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Agenda, Friday 8th, June, 2007

- General view
 - Anchor lexicon to formal representation of the world
 - Cultural perception of disjunct types => identity criteria
 - Relations between synsets and ontology
 - names
 - conceptualizations
- Simplified KIF expressions for conceptualizations
- Extend the ontology
 - New Terms (disjunct types)
 - Relations
 - Level of conceptualization:
 - afvalproduct => SUMO more abstract, more generic
 - afvalwater
- Abstract mappings from SUMO to Cornetto
 - Major structures in DWN
 - possible SUMO mappings
- SUMO/MILO axioms for constraint checking
 - constraints projected through English WordNet
 - constraints manually assigned to Dutch synsets
- Sigma
- Declarations
- START

Relation between lexical meaning & ontology

Ontology versus lexicon

- Ontology is based on logical observation within a culture => relatively independent of language
- Terms represent disjoint Types, organized in a Type hierarchy where:
 - a Type represents a class of entities that share the same essential properties.
 - Instances of a Type belong to only a single Type: => disjoint (you cannot be both a *cat* & a *dog*)
- Terms can be combined in a knowledge representation language to form expressions of axioms (*you can be a watch dog & a bull dog*)

Ontology versus lexicon

- Ontology provides an independent anchoring of the relational meaning in DWN
- Ontology can be used to validate the lexical semantic structures, both for lexical units and synsets
- Ontology provides a mapping of vocabulary to formal representations for semantic web applications
- Lexical units map onto single synsets, but synsets can be related to the ontology in various ways



Identity criteria applied to DWN

- (Semi-)rigid type hierarchy in the ontology:
 - Canine => PoodleDog; NewfoundlandDog; DalmatianDog, etc.
- Wordnet consists of names for (semi-)rigid dog-types and other words for dogs with roles:
 - poedel = PoodleDog
 - jachthond (?CAN)
 - ⇔ (and
(instance ?CAN Canine)
(instance ?EV Hunting)
(agent ?CAN ?EV)))
- Type hierarchy remains compact and pure

Ontology versus lexicon

- Synsets can be names for terms that are disjoint Types:
 - $\text{water}_{\text{NL}} = \text{Water}_{\text{SUMO}}$
 - $\text{zand}_{\text{NL}} = \text{Sand}_{\text{SUMO}}$
- Many Synsets are lexicalizations that can name instances of the same Type in different contexts:
 - water used for a purpose
 - water occurring somewhere or originating from
 - water being the result of a process
- The latter do not grant the introduction of new Types in the ontology

Comparing the DWN with SUMO

- Difficulties distinguishing the types:
 - $\{\text{water:1;majem=liquid appearance}\}_{\text{DWN}} \cong (\text{PhysicalStateFn Water Liquid})_{\text{SUMO}}$
 - $\{\text{water:2}\} = \text{WaterArea}_{\text{SUMO}}$
 - $\{\text{water:4=element}\} = \text{Water}_{\text{SUMO}} [\text{chemical element}]$
 - $\text{water [x]NL} =$
 - $(\text{instance ?W Water})$
 - $(\text{attribute ?W Liquid})$

DEBVisDic English WordNet

water

Search Search in all

[n] body of water:1, water:2
 [n] urine:1, piss:1, pee:1, piddle:1, weewee:1, water:5
 [n] water system:2, water supply:1, water:3
 [n] water:1, H2O:1
 [n] water:4

Preview Tree Revtree Edit Query Xml

Synonyms: water:1, H2O:1

Definition: binary compound that occurs at room temperature as a clear colorless odorless tasteless liquid; freezes into ice below 0 degrees centigrade and boils above 100 degrees centigrade; widely used as a solvent

Domain: chemistry

SUMO/MILO: = Water

--> [hypernym] binary compound:1
 --> [hypernym] liquid:3
 --> [holo_portion] tear:1, teardrop:2
 --> [holo_portion] perspiration:1, sweat:1, sudor:1
 --> [holo_portion] body of water:1, water:2
 --> [holo_portion] snowflake:1, flake:1
 --> [holo_portion] ice crystal:1, snow mist:1, diamond dust:1, poudrin:1, ice needle:1, frost snow:1, frost mist:1
 --> [holo_portion] dewdrop:1
 --> [holo_portion] ice:1, water ice:1
 --> [eng_derivative] water:1, irrigate:1
 --> [eng_derivative] water:2
 <<-- [eng_derivative] water:1, irrigate:1
 <<-- [eng_derivative] water:2
 <<-- [derived] aqueous:1
 <<-- [derived] aquatic:1
 <<-- [derived] hydraulic:1
 <<-- [hyponym] tap water:1
 <<-- [mero_portion] hydrogen:1, H:1, atomic number 1:1
 <<-- [mero_portion] oxygen:1, O:2, atomic number 8:1
 <<-- [hyponym] bath water:1

Querying a dictionary is complete. Item(s): 10

Water used for a purpose

- theewater (*for making tea*)
 - (instance, 0, Water) , (instance, 1, Tea) , (component, 0, 1) (instance, 2, Making) , (component, 0, 2)
 - (exists (?A ?W)
 - (and
 - (instance ?W Water)
 - (hasPurposeForAgent ?W
 - (exists (?T)
 - (and
 - (instance ?T Tea)
 - (part ?W ?T))))))
- koffiewater (*for making coffee*)
- bluswater (*for extinguishing fire*)
 - (instance, 0, Water) , (instance, 1, Fire) , (instance, 2, Process) , (instrument, 0, 2) , (finishes, 2, 1)
- scheerwater (*for shaving*)
- afwaswater (*for cleaning dishes*)
- waswater (*for washing*)
 - (instance, 0, Water) , (instance, 1, Washing) , (instrument, 0, 1)
- badwater (*for bading*)
- koelwater (*for cooling*)
- spoelwater (*for flushing*)
- drinkwater (*for drinking*)
 - (instance, 0, Water) , (instance, 1, Drinking) , (patient, 0, 1)

- theewater (*for making tea*)
 - (instance, 0, Water) , (instance, 1, Tea) , (component, 0, 1)
(instance, 2, Making) , (component, 0, 2)
 - (exists (?A ?W)
 - (and
 - (instance ?W Water)
 - (hasPurposeForAgent ?W
 - (exists (?T)
 - (and
 - (instance ?T Tea)
 - (part ?W ?T)
 -)
 -) ?A)
 -)
 -)

- bluswater (*for extinguishing fire*)
- (exists (?W ?A)
- (and
- (instance ?W Water)
- (hasPurposeForAgent ?W
- (exists (?E)
- (and
- (instance ?E Extinguishing)
- (resource ?E ?W))) ?A))
-

- drinkwater (*for drinking*)
- (exists (?W ?A)
- (and
- (instance ?W Water)
- (capability Drinking resource ?W)))

Water occurring somewhere or originating from

- ❑ putwater (*in a well*)
 - (instance, 0, Water) , (instance, 1, MineOrWell) , (origin, 0, 1)
- ❑ slootwater (*in a ditch*)
 - (instance, 0, Water) , (instance, 1, WaterArea) , (origin, 0, 1)
- ❑ oppervlaktewater (*at the surface*)
- ❑ welwater (*out of a spring*)
 - (instance, 0, Water) , (instance, 1, Creek) , (origin, 0, 1)
- ❑ leidingwater, gemeentepils, kraanwater (*out of the tap*)
 - (instance, 0, Water) , (instance, 1, Device) , (origin, 0, 1)
- ❑ gootwater (*in the kitchen sink or gutter*)
- ❑ grachtwater (*in a canal*)
 - (instance, 0, Water) , (instance, 1, Canal) , (origin, 0, 1)
- ❑ kwelwater (*coming from underneath a dike*)
 - (instance, 0, Water) , (instance, 1, StationaryArtifact) , (origin, 0, 1)
- ❑ grondwater, grondwater (*in the ground*)
- ❑ buiswater (*on a ship*)

- putwater (*in a well*)
- (exists (?W ?M)
- (and
- (instance ?W Water)
- (instance ?M MineOrWell)
- (located ?W ?M)))

- badwater (bath water)
- slootwater (*in a ditch*)
- (exists (?W ?S)
- (and
- (instance ?W Water)
- (instance ?S SmallStaticWaterArea)
- (part ?W ?S)))

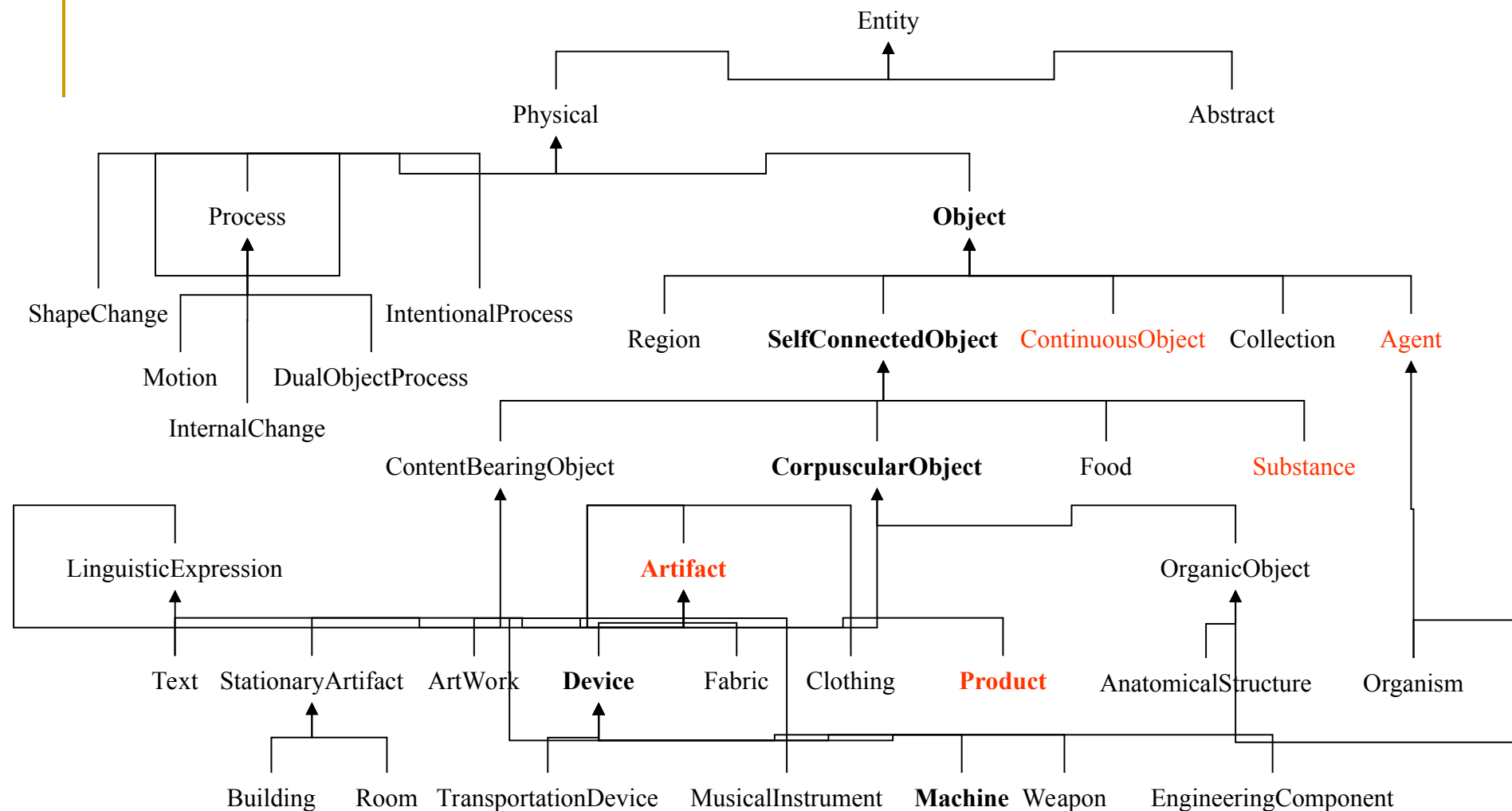
- leidingwater, gemeentepils, kraanwater (*out of the tap*)
- (exists (?W ?R ?F)
- (and
- (instance ?W Water)
- (instance ?F Faucet(=Device))
- (instance ?R Removing)
- (origin ?R ?F)
- (patient ?R ?W)))

- oppervlaktewater (*at the surface*) = WaterArea
- grondwater (under the ground) = GroundWater
- WaterArea disjoint GroundWater

Being the result of a process

- ❑ pompwater (*being pumped away*)
 - (instance, 0, Water) , (instance, 1, Pump) , (instance, 2, Transportation) , (instrument, 1, 2) , (patient, 0, 2)
- ❑ smeltwater, dooiwater (*melting snow and ice*)
 - (instance, 0, Water) , (instance, 1, Melting) , (result, 0, 1)
- ❑ afvalwater (*waste water*)
- ❑ condens, condensatiewater, condenswater (*from condensation*)
- ❑ lekwater (*leaking water*)
- ❑ regenwater (*rain water*)
 - (instance, 0, Water) , (instance, 1, Raining) , (result, 0, 1)
- ❑ spuiwater (*being drained for water maintenance*)





1st	HAS_HYPONYM afscheiding 2 NODE 19 / 232 (separation)
1st	HAS_HYPONYM bron 2 NODE 12 / 126 (source)
1st;2nd;3rd	HAS_HYPONYM deel 2 NODE 375 / 5691 (part)
2nd	HAS_HYPONYM eigenschap 1 NODE 103 / 4231 (property)
2nd	HAS_HYPONYM gebeurtenis 1 NODE 52 / 13198 (event,
NOUN)	
3rd	HAS_HYPONYM gedachte 2 NODE 30 / 1368 (thought)
1st;2nd;3rd	HAS_HYPONYM geheel 1 NODE 29 / 629 (whole)
1st;2nd;3rd	HAS_HYPONYM groep 4 NODE 102 / 5923 (group)
1st	HAS_HYPONYM hoeveelheid 1 NODE 128 / 768 (quantity)
1st;2nd;3rd	HAS_HYPONYM middel 2 NODE 50 / 5266 (means)
1st	HAS_HYPONYM object 1 NODE 4 / 17986 (object)
1st	HAS_HYPONYM plaats 1 NODE 92 / 4041 (place)
1st	HAS_HYPONYM produkt 1 NODE 48 / 2469 (product)
1st	HAS_HYPONYM samenstelling 3 NODE 1 / 441
(composition)	
1st	HAS_HYPONYM soort 2 NODE 29 / 227 (kind)
1st	HAS_HYPONYM substantie 1 NODE 89 / 4993 (substance)
2nd	HAS_HYPONYM tijd 1 NODE 10 / 824 (time)
2nd	HAS_HYPONYM toestand 1 NODE 123 / 4329 (situation,
NOUN)	
1st;2nd	HAS_HYPONYM uiting 2 NODE 41 / 5975 (utterance)
2nd	XPOS_HAS_HYPONYM gebeuren 2 NODE (to happen,
VERB)	
2nd	XPOS_HAS_HYPONYM zijn 7 NODE (to be, VERB)

_gebeuren 2

HAS_HYPONYM beginnen 3 NODE 8 / 12

HAS_HYPONYM doorgaan 4 NODE 3 / 4

HAS_HYPONYM handelen 2 NODE 53 / 4429 (to act)

HAS_HYPONYM meemaken 1 NODE 9 / 256 (to experience)

HAS_HYPONYM missen 4 NODE 2 / 2

HAS_HYPONYM plaatsvinden 1 NODE 4 / 5

HAS_HYPONYM samenvallen 2 NODE 1 / 1

HAS_HYPONYM stoppen 8 NODE 10 / 21

HAS_HYPONYM uitoefenen 2 NODE 4 / 421 (to excel force)

HAS_HYPONYM veranderen 1 NODE 154 / 2358 (to change)

HAS_HYPONYM verliezen 1 NODE 8 / 9

HAS_HYPONYM verlopen 4 NODE 2 / 2

HAS_HYPONYM veroorzaken 1 NODE 48 / 6242 (to cause)

HAS_HYPONYM vinden 1 NODE 5 / 11

HAS_HYPONYM voorkomen 4 NODE 6 / 10

HAS_HYPONYM zullen 2 NODE 1 / 1

Dynamic cluster “gebeuren” (to happen)

- ❑ causal change:
 - veroorzaken (to cause);
- ❑ non-causal change:
 - veranderen (to change as a process)
- ❑ phenomena:
 - voorkomen (to occur); plaatsvinden (to take place); overkomen (to happen)
- ❑ actions:
 - handelen (to act)
- ❑ experience:
 - meemaken (to undergo, experience)
- ❑ exert force:
 - uitoefenen (to exert force)
- ❑ aspectuals:
 - beginnen (to begin); doorgaan (to continue); stoppen (to stop); herhalen (to repeat); samenvallen (to co-occur);

Multiple hyperonyms

- Regular compositional verb structures:
 - doorademen:
 - door+ademen (lit. through+breath, continue to breath)
 - doorbetalen:
 - door+betalen (lit. through+pay, continue to pay)
 - doorlopen:
 - door+lopen (lit. through+walk, continue to walk)
 - doorfietsen:
 - door+fietsen (lit. through+walk, continue to walk)
 - doorrijden:
 - door+rijden (lit. through+walk, continue to walk)

Adjective+verb compounds

- openknijpen:
 - open+knijpen (lit. open+squeeze, to open by squeezing)
- opendraaien:
 - open+draaien (lit. open+turn, to open by turning)
- openschuiven:
 - open+schuiven (lit. open+slide, to open by sliding)
- dichtknijpen:
 - dicht+knijpen (lit. closed+squeeze, to close by squeezing)
- dichtdraaien:
 - dicht+draaien (lit. closed +turn, to close by turning)
- dichtschuiven:
 - dicht+schuiven (lit. closed +slide, to close by sliding)
- MORE GENERAL VERBS:
- openmaken:
 - open+maken (lit. open+make, to cause to be open);
- dichtmaken:
 - dicht+maken (lit. dicht+make, to cause to be open);

Adjective+verb compounds

- MORE GENERAL VERBS:
- openmaken:
 - open+maken (lit. open+make, to cause to be open);
- dichtmaken:
 - dicht+maken (lit. dicht+make, to cause to be open);
- MORE SPECIFIC VERBS:
- openknijpen: open+knijpen (lit. open+squeeze, to open by squeezing)
 - has_hyperonym knijpen (squeeze)
 - has_hyperonym openmaken (to open)
- opendraaien: open+draaien (lit. open+turn, to open by turning)
 - has_hyperonym draaien (to turn)
 - has_hyperonym openmaken (to open)
- dichtknijpen: dicht+knijpen (lit. closed+squeeze, to close by squeezing)
 - has_hyperonym knijpen (squeeze)
 - has_hyperonym dichtmaken (to close)
- dichtdraaien: dicht+draaien (lit. closed +turn, to close by turning)
 - has_hyperonym draaien (to turn)
 - has_hyperonym dichtmaken (to close)



Parallel Noun and Verb hierarchy

- dynamic nodes
 - NOUN gebeurtenis (event)
 - VERB gebeuren (to happen))
- static nodes:
 - NOUN toestand (situation)
 - NOUN eigenschap (property)
 - VERB zijn (to be)

Nominal Hierarchy					Verbal Hierarchy			
gebeurtenis (event)				<XPOS_SYN>				gebeuren (to happen)
	handeling (an act)			<XPOS_SYN>			handelen (to act)	
		daad (deed)		<XPOS_SYN>		doen (to do)		
	verandering (a change)			<XPOS_SYN>			veranderen (to change, intrans)	
		beweging (movement)		<XPOS_SYN>		bewegen (to move, intrans)		
			verplaatsing (change of location)	<XPOS_SYN>	verplaatsen (to move position, intrans)			
				<XPOS_SYN>			veroorzaken (to cause)	
	verandering (a change)			<XPOS_SYN>		veranderen (to change, trans)		
		beweging (movement)		<XPOS_SYN>	bewegen (to move, trans)			

_ zijn 7

HAS_HYPONYM bevinden 2 NODE 27 / 255

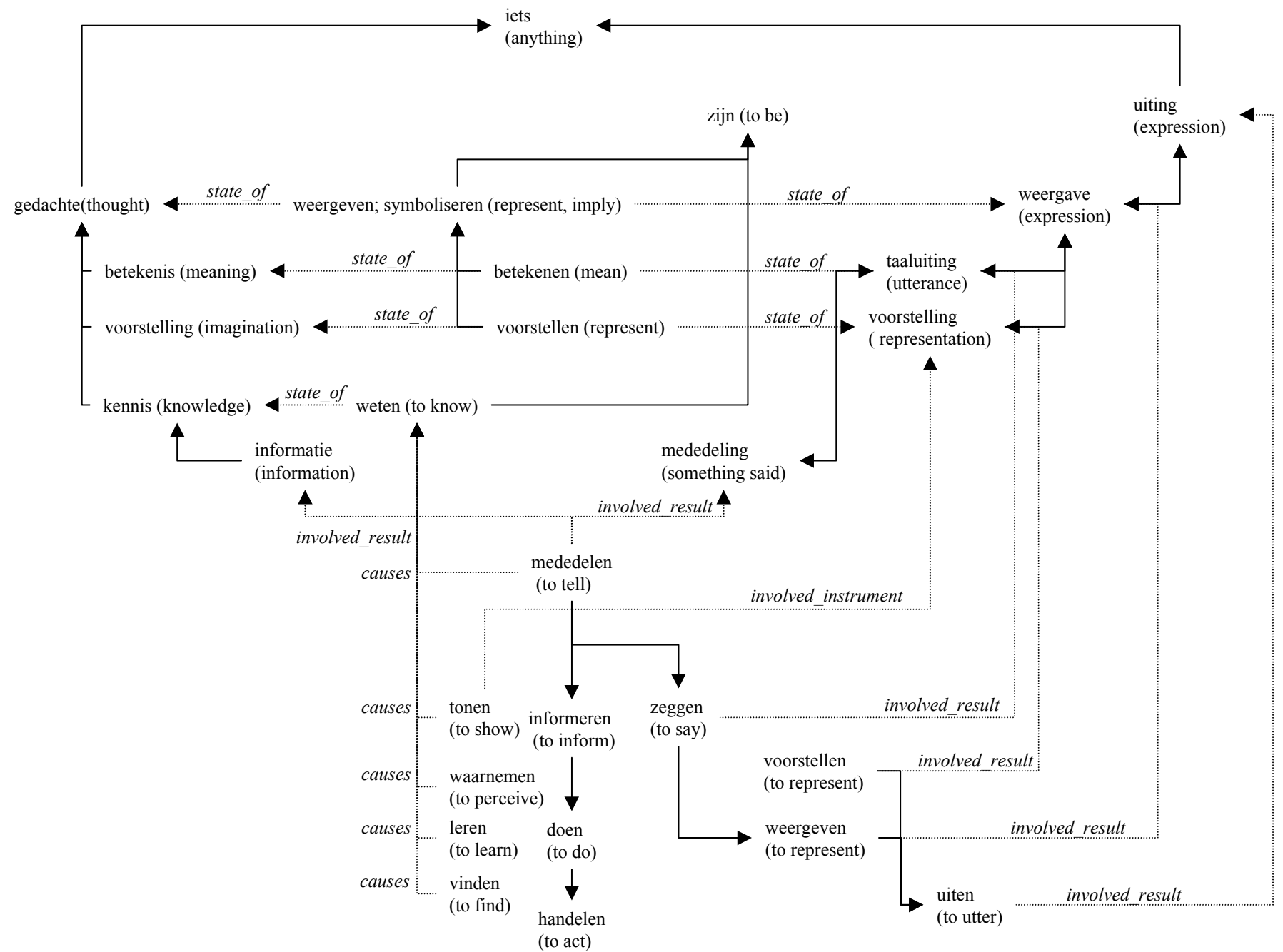
HAS_HYPONYM denken 4 NODE 9 / 53

HAS_HYPONYM eruitzien 1 NODE 13 / 39

HAS_HYPONYM zijn 2 NODE 5 / 21

HAS_HYPONYM zullen 2 NODE 1 / 1

- “kleur” (colour)
- has_xpos_hyponym: zwart (black), wit (white); rood (red); geel (yellow); blauw (blue); groen (green); paars (purple); oranje (orange)



Constraint violations

- ❑ LUs/Synsets that violate constraints directly
 - Within the same information type
 - Across information types
- ❑ LUs/Synsets that violate constraints through inheritance
 - hyponymy
 - antonymy
 - meronymy
 - ontology: types & roles; instances

